

BHUVANESHWARI D

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE student at SASTRA Deemed University with hands-on experience in data analytics, ML-powered decision intelligence, and AI-driven business insights. Proficient in Python, SQL, Scikit-learn, Pandas, and LangChain. Built end-to-end analytics systems delivering actionable recommendations for retail and healthcare domains. Strong at translating complex business problems into scalable AI solutions and communicating insights to both technical and non-technical stakeholders.

EDUCATION

B.Tech Computer Science and Engineering | CGPA: 7.2 **2027**

SASTRA Deemed University, Kumbakonam

Higher Secondary Education | 86.5% **2023**

Christ The King Girls Matriculation Higher Secondary School

TECHNICAL SKILLS

- Languages** : Python, C++, Java, SQL
- AI / ML** : PyTorch, TensorFlow, Scikit-learn, LangChain, NLP, Computer Vision
- Deep Learning** : CNN, U-Net, ResNet-50, SE Attention, CBAM, Mamba Architecture
- Agentic AI** : LangChain, Multi-step AI Orchestration, Agentic Pipeline Design
- Data & Analytics** : Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Statistical Analysis
- Tools** : Git, GitHub, Streamlit, Postman, Jupyter Notebook, Kaggle, VS Code

PROJECTS & EXPERIENCE

AK-Net– Attention-Based Feature Fusion for Medical Segmentation

PyTorch | ResNet-50 | SE Attention | CBAM | Mamba | Deep Learning **2026**

- Designed AK-Net using ResNet-50 encoder with SE and CBAM dual-attention mechanisms, achieving improved segmentation performance over baseline U-Net across 4 medical datasets — Synapse, CVC-ClinicDB and ISIC-2017.
- Implemented Mamba bottleneck for long-range dependency modeling and contributed to research paper covering architecture design, experimental analysis, and results for journal submission.

RetailSense Agent – Agentic AI for Smart Retail Decision Intelligence

Python | LangChain | LLM | Flask | RESTful API | Agentic Pipeline **2025**

- Built agentic AI web application using LangChain and LLMs to automate retail demand analysis, inventory insights, and business recommendations — eliminating manual decision-making effort for retail operations.
- Designed multi-step AI orchestration pipeline processing retail data end-to-end, delivering natural language business insights aligned with enterprise decision intelligence use cases.

Diabetes Risk Intelligence System

Python | Scikit-learn | K-Means | Pandas | Streamlit | Matplotlib | ReportLab **2025**

- Processed 768 patient records with end-to-end ML pipeline — handling missing clinical values via median imputation and engineering custom Risk Scoring engine ($\text{Glucose} \times 0.4 + \text{BMI} \times 0.3 + \text{Age} \times 0.3$) to classify patients into 3 risk groups with scores normalized to 0–100.
- Deployed real-time Streamlit dashboard with K-Means patient segmentation, anomaly alerts, personalized health recommendations, and one-click downloadable PDF reports — enabling last-mile clinical decision support.

SOFT SKILLS

Independent Work Capability | Team Collaboration | Adaptability | Analytical Thinking | Client Communication | Excellence-first Mindset

LANGUAGES

Tamil (Native) | English (Professional)